

CAPSTONE PROJECT

Business Analytics & Performance Analysis

**An End-to-End Data Processing & BI
Solution**

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Submitted To:

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Section 1: Data Exploration & Initial Analysis (Excel)

1.1 Project Overview & Data Ingestion:

Name	Type	Compressed size
 categories	Microsoft Excel Comma S...	1 KB
 customers	Microsoft Excel Comma S...	6 KB
 employee_territory	Microsoft Excel Comma S...	1 KB
 employees	Microsoft Excel Comma S...	1 KB
 order_details	Microsoft Excel Comma S...	12 KB
 orders	Microsoft Excel Comma S...	24 KB
 products	Microsoft Excel Comma S...	3 KB
 region	Microsoft Excel Comma S...	1 KB
 shippers	Microsoft Excel Comma S...	1 KB
 suppliers	Microsoft Excel Comma S...	3 KB
 territories	Microsoft Excel Comma S...	1 KB

Technical Documentation:

The project utilizes the Northwind Database, consisting of 11 core relational tables. These tables were ingested to perform comprehensive data preprocessing, transformation, and exploratory data analysis.

1.2 Data Preparation & Relational Calculations

A	B	C	D	E	F	G	H	I	J	K	L	M
OrderID	Product Name	product_name_upper	Full product details	ProductID	UnitPrice	Quantity	Discount	Sales Amount	Order size	order date	Customer ID	Ship Country
10248	Queso Cabrales	QUESO CABRALES		11	\$14.00	12	0.0%	\$168.00	Medium	7/4/1996	VINET	France
10248	Singaporean Hokkien Fried Mee	SINGAPOREAN HOKKIEN FRIED MEE	42_Singaporean Hokkien Fried Mee	42	\$9.80	10	0.0%	\$98.00	Small	7/4/1996	VINET	France
10248	Mozzarella di Giovanni	MOZZARELLA DI GIOVANNI	72_Mozzarella di Giovanni	72	\$34.80	5	0.0%	\$174.00	Small	7/4/1996	VINET	France
10249	Tofu	TOFU	14_Tofu	14	\$18.60	9	0.0%	\$167.40	Small	7/5/1996	TOMSP	Germany
10249	Manjimup Dried Apples	MANJIMUP DRIED APPLES	51_Manjimup Dried Apples	51	\$42.40	40	0.0%	\$1,696.00	Large	7/5/1996	TOMSP	Germany
10250	Jack's New England Clam Chowder	JACK'S NEW ENGLAND CLAM CHOW	41_Jack's New England Clam Chowd	41	\$7.70	10	0.0%	\$77.00	Small	7/8/1996	HANAR	Brazil
10250	Manjimup Dried Apples	MANJIMUP DRIED APPLES	51_Manjimup Dried Apples	51	\$42.40	35	0.0%	\$1,484.00	Large	7/8/1996	HANAR	Brazil
10250	Louisiana Fiery Hot Pepper Sauce	LOUISIANA FIERY HOT PEPPER SAUC	65_Louisiana Fiery Hot Pepper Sauci	65	\$16.80	15	0.0%	\$252.00	Medium	7/8/1996	HANAR	Brazil
10251	Gustaf's Knckebrd	GUSTAF'S KNCKEBRD	22_Gustaf's Knckebrd	22	\$16.80	6	0.0%	\$100.80	Small	7/8/1996	VICTE	France
10251	Ravioli Angelo	RAVIOLI ANGELO	57_Ravioli Angelo	57	\$15.60	15	0.0%	\$234.00	Medium	7/8/1996	VICTE	France
10251	Louisiana Fiery Hot Pepper Sauce	LOUISIANA FIERY HOT PEPPER SAUC	65_Louisiana Fiery Hot Pepper Sauci	65	\$16.80	20	0.0%	\$336.00	Medium	7/8/1996	VICTE	France
10252	Sir Rodney's Marmalade	SIR RODNEY'S MARMALADE	20_Sir Rodney's Marmalade	20	\$64.80	40	0.0%	\$2,592.00	Large	7/9/1996	SUPRD	Belgium
10252	Geitost	GEITOST	33_Geitost	33	\$2.00	25	0.0%	\$50.00	Medium	7/9/1996	SUPRD	Belgium
10252	Camembert Pierrot	CAMEMBERT PIERROT	60_Camembert Pierrot	60	\$27.20	40	0.0%	\$1,088.00	Large	7/9/1996	SUPRD	Belgium
10253	Gorgonzola Telino	GORGONZOLA TELINO	31_Gorgonzola Telino	31	\$10.00	20	0.0%	\$200.00	Medium	7/10/1996	HANAR	Brazil
10253	Chartreuse verte	CHARTREUSE VERTE	39_Chartreuse verte	39	\$14.40	42	0.0%	\$604.80	Large	7/10/1996	HANAR	Brazil
10253	Maxilaku	MAXILAKU	49_Maxilaku	49	\$16.00	40	0.0%	\$640.00	Large	7/10/1996	HANAR	Brazil
10254	Guaran Fantstica	GUARAN FANTSTICA	24_Guaran Fantstica	24	\$3.60	15	0.0%	\$54.00	Medium	7/11/1996	CHOPS	Switzerland
10254	Pt chinois	PT CHINOIS	55_Pt chinois	55	\$19.20	21	0.0%	\$403.20	Medium	7/11/1996	CHOPS	Switzerland
10254	Longlife Tofu	LONGLIFE TOFU	74_Longlife Tofu	74	\$8.00	21	0.0%	\$168.00	Medium	7/11/1996	CHOPS	Switzerland
10255	Chang	CHANG	2_Chang	2	\$15.20	20	0.0%	\$304.00	Medium	7/12/1996	RICSU	Switzerland
10255	Pavlova	PAVLOVA	16_Pavlova	16	\$13.90	35	0.0%	\$486.50	Large	7/12/1996	RICSU	Switzerland
10255	Inlagd Sill	INLAGD SILL	36_Inlagd Sill	36	\$15.20	25	0.0%	\$380.00	Medium	7/12/1996	RICSU	Switzerland
10255	Raclette Courdavault	RACLETTE COURDAVAULT	59_Raclette Courdavault	59	\$44.00	30	0.0%	\$1,320.00	Medium	7/12/1996	RICSU	Switzerland
10256	Perth Pasties	PERTH PASTIES	53_Perth Pasties	53	\$26.20	15	0.0%	\$393.00	Medium	7/15/1996	WELLI	Brazil
10256	Original Frankfurter grne Soe	ORIGINAL FRANKFURTER GRNE SOE	77_Original Frankfurter grne Soe	77	\$10.40	12	0.0%	\$124.80	Medium	7/15/1996	WELLI	Brazil
10257	Schoggi Schokolade	SCHOGGI SCHOKOLADE	27_Schoggi Schokolade	27	\$35.10	25	0.0%	\$877.50	Medium	7/16/1996	HILAA	Venezuela
10257	Chartreuse verte	CHARTREUSE VERTE	39_Chartreuse verte	39	\$14.40	6	0.0%	\$86.40	Small	7/16/1996	HILAA	Venezuela

Technical Documentation:

Data Cleansing & Feature Engineering:

The project began with data hygiene across all tables by handling missing values, eliminating duplicates, and standardizing data types. We enhanced the orders table by calculating logistics efficiency (shipping_duration = shipped_date - order_date) and extracting temporal dimensions (Order Year and Order Month). Furthermore, the order_details dataset was enriched using advanced VLOOKUP integrations to fetch product names, customer details, and shipping countries, alongside custom text formatting (UPPER and concatenation for full product details).

1.3 Pivot Tables & Exploratory Analysis:

order date	Sum of Sales Amount	product name	Sum of Sales Amount	Sum of Quantity	Sum of Sales Amount	Sum of Quantity	Count of OrderID
1996		Cte de Blaye	\$149,984.20	623	\$1,354,458.59	51317	2155
Qtr3		Thuringer Rostbratwurst	\$87,736.40	746			
Jul	\$30,192.10	Raclette Courdavault	\$76,236.00	1496			
Aug	\$26,603.40	Camembert Pierrot	\$50,286.00	1577			
Sep	\$27,636.00	Tarte au sucre	\$43,827.90	1083			
Qtr4		Gnocchi di nonna Alice	\$45,121.20	1263			
Oct	\$41,203.60	Manjimup Dried Apples	\$44,742.60	886			
Nov	\$49,704.00	Alice Mutton	\$35,482.20	978			
Dec	\$50,353.40	Cornarvon Tigers	\$31,987.50	539			
1997		Rozle Sauerbrout	\$26,855.60	640			
Qtr1	\$147,879.90	Grand Total	\$598,329.60	9831			
Qtr2	\$151,611.09						
Qtr3							
Jul	\$55,464.93						
Aug	\$49,981.69						
Sep	\$59,733.02						
Qtr4							
Oct	\$70,328.50						
Nov	\$45,913.36						
Dec	\$77,476.26						
1998							
Qtr1	\$315,242.12						
Qtr2	\$154,529.22						
Grand Tot:	\$1,354,458.59						

ship country	Sum of Sales Amount
USA	\$263,566.98
Germany	\$244,640.63
Austria	\$139,436.63
Brazil	\$114,968.48
France	\$85,498.76
Grand Total	\$848,171.48

Technical Documentation:

To aggregate and summarize the massive relational data efficiently, Excel Pivot Tables and Pivot Charts were built. This allowed us to quickly group sales by categories, track performance trends across years, and analyze top-performing products and regions prior to final visualization.

1.4 The Final Dashboard & Visualizations



Technical Documentation:

Interactive Sales Performance Dashboard:

The final output is a professional, dark-themed Northwind Sales Performance Dashboard. It features custom KPI Cards linked dynamically via Excel shapes, visual charts including a yearly Sales Trend line chart, Top 10 Products, and regional distributions. The dashboard incorporates interactive Slicers (such as Customer ID and Ship Country) to enable seamless, multi-dimensional business filtering and performance tracking.

Section 2: Data Preparation & Modeling (Power Query & Power Pivot)

2.1 Data Cleaning & Transformation (Power Query)

Table: ReplaceValue(#"Replaced Value1",null,"unknown",Replacer.ReplaceValue,{"CompanyName"})

	ShipCity	ShipRegion	ShipCountry	CompanyName	days to ship	order_size
1	Reims	unknown	France	Vins et alcools Chevalier	12	medium freight
2	Münster	unknown	Germany	Toms Spezialitäten	5	low freight
3	Lyon	unknown	France	Victuailles en stock	7	medium freight
4	Rio de Janeiro	RJ	Brazil	Hanari Carnes	4	high freight
5	Charleroi	unknown	Belgium	Suprêmes délices	2	high freight
6	Rio de Janeiro	RJ	Brazil	Hanari Carnes	6	high freight
7	Bern	unknown	Switzerland	Chop-suey Chinese	12	medium freight
8	Genève	unknown	Switzerland	Richter Supermarkt	3	high freight
9	Resende	SP	Brazil	Wellington Importadora	2	low freight
10	San Cristóbal	Tchira	Venezuela	HILARION-Abastos	6	high freight
11	Graz	unknown	Austria	unknown	6	high freight
12	México D.F.	unknown	Mexico	Centro comercial Moctezuma	7	low freight
13	Rio de Janeiro	RJ	Brazil	Que Pasa	11	low freight
14	Klin	unknown	Germany	Ottlies Kesseladen	10	high freight
15	Albuquerque	NM	USA	Rattlesnake Canyon Grocery	3	medium freight
16	Graz	unknown	Austria	unknown	8	high freight
17	Brücke	unknown	Sweden	Folk och f HB	30	low freight
18	Strasbourg	unknown	France	Blondel's pre et fils	18	high freight
19	Oulu	unknown	Finland	Wartian Herkku	5	medium freight
20	München	unknown	Germany	Frankenversand	8	high freight
21	Caracas	DF	Venezuela	GROSELLA-Restaurante	3	high freight
22	Seattle	WA	USA	White Clover Markets	9	low freight
23	Lander	WY	USA	Split Rail Beer & Ale	29	low freight
24	Oulu	unknown	Finland	Wartian Herkku	1	high freight
25	Albuquerque	NM	USA	Rattlesnake Canyon Grocery	4	high freight

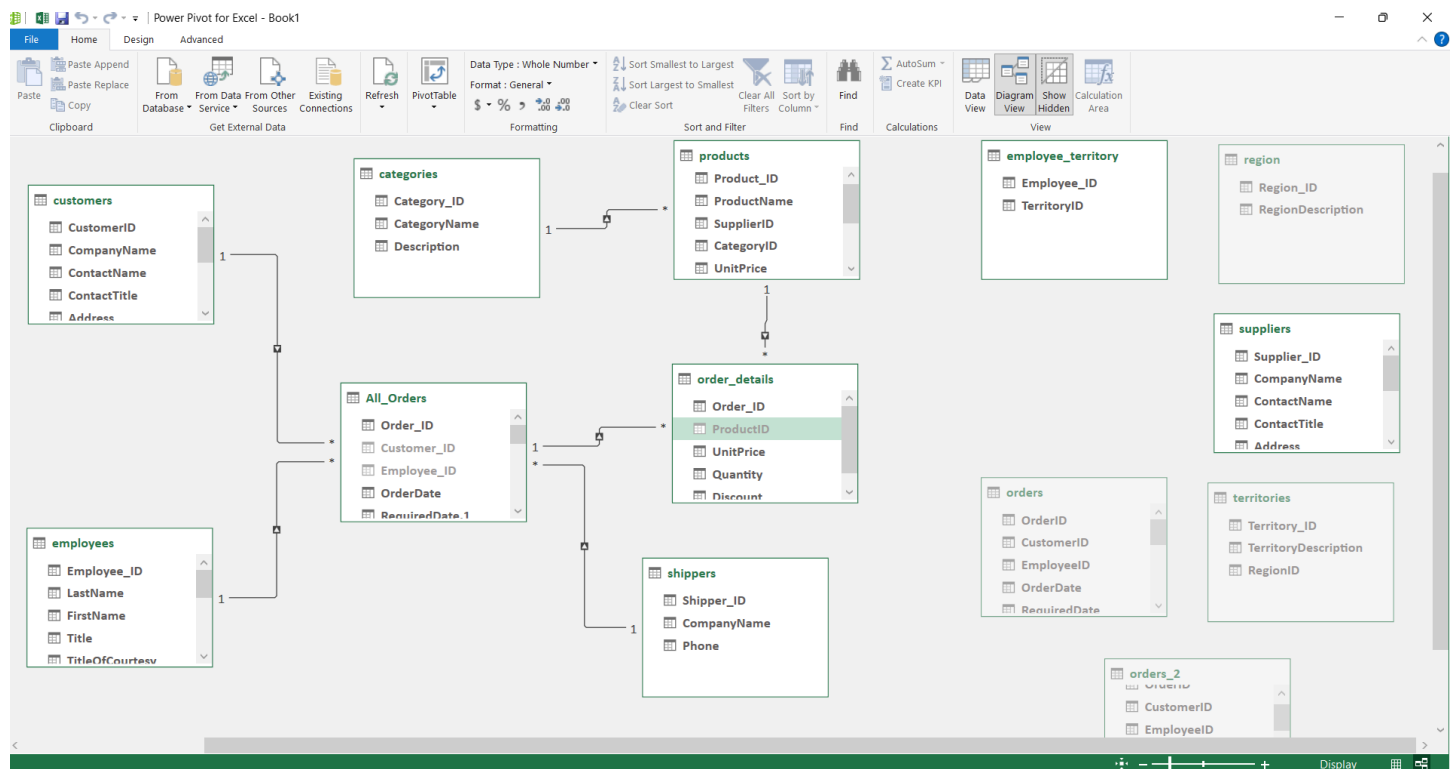
Technical Documentation:

To establish an automated and reusable data preparation workflow, raw datasets were imported from multiple source formats, including **Excel** (containing 1996 & 1997 data) and **CSV** (containing 1998 data). The following transformations were executed inside Power Query:

- **Data Cleaning & Handling Missing Values:** Inspected column quality, identified missing values, and replaced NULL values in the ShipRegion attribute with "unknown" using the Replace Values transformation.

- **Column Transformations:** Adjusted data types across all columns, renamed headers for consistency, removed unnecessary columns, and eliminated duplicate rows (Remove Duplicates).
- **Column Splitting & Sorting:** Applied text splitting (Split Column by Delimiter) to extract specific text attributes and sorted rows to streamline data structure.
- **Merging Queries:** Applied Merge Queries to combine lookup tables, retrieving customer details (Customer Name) and carrier details (CompanyName) into the unified table.
- **Calculated & Conditional Columns:**
 - Created a custom calculated column **days to ship** by subtracting OrderDate from ShippedDate (Inserted Date Subtraction).
 - Created a **Conditional Column** named **order_size** based on freight costs:
 - If Freight > 50 \$ \rightarrow\$ "high freight"
 - Elself Freight > 40 \$ \rightarrow\$ "medium freight"
 - Else \$ \rightarrow\$ "low freight"
- **Load to Data Model:** The fully cleaned and transformed dataset (All_Orders) was loaded directly into the **Power Pivot Data Model** via Close & Load To....

2.2 Data Modeling & Relationships (Power Pivot Diagram View)



Data Model Architecture: Built a structured **Star Schema** data model by establishing directional relationships (1:*) between the central Fact tables (All_Orders / order_details) and surrounding Dimension tables (customers, employees, products, shippers, categories).

Model Optimization: Hid unneeded tables and system columns from the client view using Hide from Client Tools / Hide from Data Model to optimize model clarity and memory efficiency.

2.3 DAX Calculations & Related Datasets (Power Pivot Data View)

Power Pivot for Excel - Book1

File Home Design Advanced

Clipboard: Paste, Paste Append, Paste Replace, Copy

Get External Data: From Database, From Data Service, From Other Sources, Existing Connections, Refresh, PivotTable

Formatting: Data Type: General, Format: General, \$, %, >, <, 0, 00, 0.00

Sort and Filter: Sort A to Z, Sort Z to A, Clear All Filters, Clear Sort, Sort by Column

Find: Find

Calculations: AutoSum, Create KPI

View: Data View, Diagram View, Show Hidden, Calculation Area

total orders:=COUNTROWS(All_Orders)

Add Column	Customer Name	order_size	days to ship	CompanyName	ShipCountry	ShipRegion	ShipCity	ShipName	Freight	ShippedDate	RequiredDate	RequiredDate
	Toms Speziallitten	low freight	5	Toms Speziallitten	Germany	NULL	Mnster	Toms Spezia...	11.61	1 ... 1 - 7 / 1996	12 ... 1 - 7 / 1996	08
	Ottilies Kseladen	high freight	10	Ottilies Kseladen	Germany	NULL	Kln	Ottilies Ksel...	55.09	1 ... 1 - 29 / 1996	12 ... 1 - 29 / 1996	08
	Frankenversand	high freight	8	Frankenversand	Germany	NULL	Mnchen	Frankenvers...	208.58	1 ... 1 - 7 / 1996	12 ... 1 - 7 / 1996	08
	QUICK-Stop	high freight	7	QUICK-Stop	Germany	NULL	Cunewalde	QUICK-Stop	76.07	3 ... 1 - 2 / 1996	12 ... 1 - 2 / 1996	09
	Morgenstern Ges...	high freight	4	Morgenstern Ges...	Germany	NULL	Leipzig	Morgenster...	125.77	3 ... 1 - 2 / 1996	12 ... 1 - 2 / 1996	09
	Lehmanns Marks...	medium freight	3	Lehmanns Markt...	Germany	NULL	Frankfurt ...	Lehmanns ...	25.83	2 ... 1 - 16 / 1996	12 ... 1 - 16 / 1996	09
	Lehmanns Marks...	high freight	8	Lehmanns Markt...	Germany	NULL	Frankfurt ...	Lehmanns ...	76.56	1 ... 1 - 27 / 1996	12 ... 1 - 27 / 1996	09
	QUICK-Stop	high freight	6	QUICK-Stop	Germany	NULL	Cunewalde	QUICK-Stop	76.83	2 ... 1 - 26 / 1996	12 ... 1 - 26 / 1996	09
	QUICK-Stop	high freight	9	QUICK-Stop	Germany	NULL	Cunewalde	QUICK-Stop	229.24	3 ... 1 - 3 / 1996	12 ... 1 - 3 / 1996	09
	Die Wandernde Kuh	medium freight	8	Die Wandernde K...	Germany	NULL	Stuttgart	Die Wander...	45.08	2 ... 1 - 17 / 1996	12 ... 1 - 17 / 1996	10
	Die Wandernde Kuh	medium freight	10	Die Wandernde K...	Germany	NULL	Stuttgart	Die Wander...	40.26	2 ... 1 - 2 / 1996	12 ... 1 - 2 / 1996	10
	QUICK-Stop	low freight	10	QUICK-Stop	Germany	NULL	Cunewalde	QUICK-Stop	1.96	2 ... 1 - 4 / 1996	12 ... 1 - 4 / 1996	10
	Kniglich Essen	low freight	7	Kniglich Essen	Germany	NULL	Brandenb...	Kniglich Essen	4.88	1 ... 1 - 14 / 1996	12 ... 1 - 14 / 1996	11
	Kniglich Essen	high freight	5	Kniglich Essen	Germany	NULL	Brandenb...	Kniglich Essen	64.86	3 ... 1 - 14 / 1996	12 ... 1 - 14 / 1996	10
	Frankenversand	high freight	5	Frankenversand	Germany	NULL	Mnchen	Frankenvers...	108.26	3 ... 1 - 29 / 1996	12 ... 1 - 29 / 1996	11
	Frankenversand	high freight	5	Frankenversand	Germany	NULL	Mnchen	Frankenvers...	54.83	2 ... 1 - 4 / 1996	12 ... 1 - 4 / 1996	11
	Lehmanns Marks...	high freight	6	Lehmanns Markt...	Germany	NULL	Frankfurt ...	Lehmanns ...	110.37	1 ... 1 - 7 / 1996	12 ... 1 - 7 / 1996	11
	QUICK-Stop	high freight	7	QUICK-Stop	Germany	NULL	Cunewalde	QUICK-Stop	249.06	2 ... 1 - 11 / 1996	12 ... 1 - 11 / 1996	12
	Die Wandernde Kuh	low freight	8	Die Wandernde K...	Germany	NULL	Stuttgart	Die Wander...	0.78	2 ... 1 - 15 / 1996	12 ... 1 - 15 / 1996	12
	Die Wandernde Kuh	medium freight	9	Die Wandernde K...	Germany	NULL	Stuttgart	Die Wander...	36.71	2 ... 1 - 27 / 1996	12 ... 1 - 27 / 1996	12
	QUICK-Stop	high freight	11	QUICK-Stop	Germany	NULL	Cunewalde	QUICK-Stop	183.17	2 ... 1 - 2 / 1996	12 ... 1 - 2 / 1996	12

total orders: ...

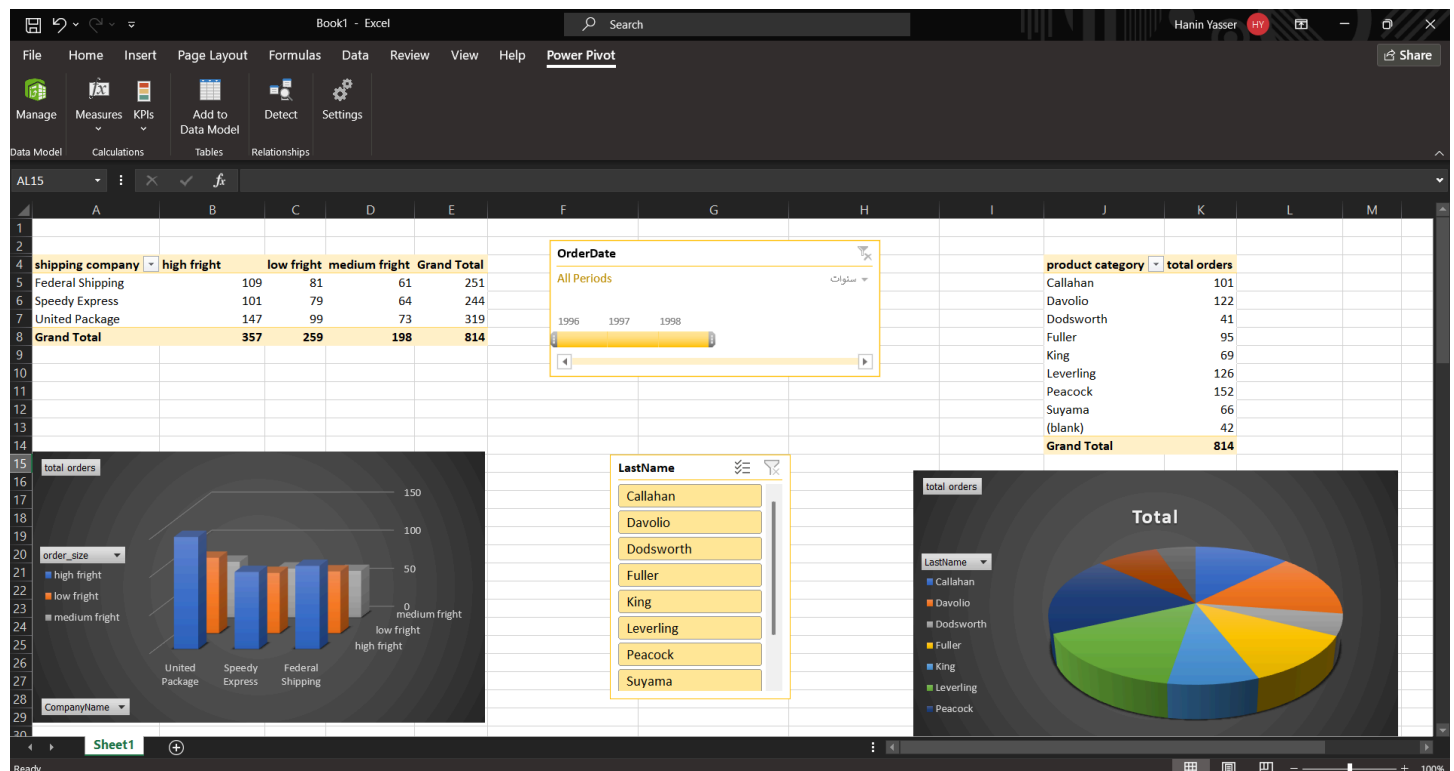
All_Orders | orders_2 | territories | suppliers | shippers | region | orders | products | order_details | employees | employee_territory | customers | categories

Record: 1 of 814

Technical Documentation:

- **Calculated Column with RELATED():** Added a calculated column to retrieve customer details from an external table using DAX:
 - Code snippet
 - Customer Name = RELATED(customers[CompanyName])
 - This demonstrates analyzing related datasets via existing model relationships.
- **Explicit DAX Measure:** Created an initial aggregation measure in the Calculation Area to track order volume:
 - Code snippet
 - total orders := COUNTROWS(All_Orders)

2.4 Initial Analytical Dashboard (Excel Pivot Tables & Charts)



Technical Documentation:

Explored initial business patterns by creating an interactive analysis dashboard driven by the Power Pivot Data Model:

- **Logistics & Freight Analysis:** Analyzed shipment distribution across carriers (shipping company) relative to shipping volume sizes (order_size categories: high, medium, low) using a Clustered Column Chart.
- **Product Category Performance:** Evaluated order counts across product categories and employee coverage using a Dynamic Pie Chart.

- **Interactive Filtering:** Integrated interactive **Slicers** (LastName for sales representatives) and a **Timeline Slicer** (OrderDate) connected across all Pivot Visuals for real-time comparative analysis.

Section 3:SQL – Business Data Analysis

3.1 Data Ingestion & Setup

Results		Messages	
	ForeignKeyName	ParentTable	ReferencedTable
1	FK_OrderDetails_Orders	order details clean	orders clean done
2	FK_Orders_Customer	orders clean done	customer clean
3	FK_OrderDetails_Product	order details clean	product_clean
4	FK_Product_Category	product_clean	category clean

- "The 5 core business tables (Customers, Orders, Order Details, Products, and Categories) were successfully ingested individually. Data types were adjusted to resolve formatting constraints, and relational integrity was enforced by establishing Primary Keys and Foreign Keys to link all transactions seamlessly."

3.2 Business Analysis & Query Results

Question 1: What are the best-performing products?

- **SQL Techniques:** INNER JOIN, GROUP BY, SUM, ORDER BY
- **Description:** Evaluated top-selling products based on total units sold and total revenue.

Results		Messages		
	ProductID	ProductName	TotalUnitsSold	TotalRevenue
1	38	Cte de Blaye	623	149984.20
2	29	Thringer Rostbratwurst	746	87736.40
3	59	Raclette Courdavault	1496	76296.00
4	60	Camembert Pierrot	1577	50286.00
5	62	Tarte au sucre	1083	49827.90
6	56	Gnocchi di nonna Alice	1263	45121.20
7	51	Manjimup Dried Apples	886	44742.60
8	17	Alice Mutton	978	35482.20
9	18	Carnarvon Tigers	539	31987.50
10	28	Rssle Sauerkraut	640	26865.60

Question 2: Which customers generate the highest revenue?

- **SQL Techniques:** Multi-table INNER JOIN, Aggregate Functions, GROUP BY, ORDER BY
- **Description:** Identified high-value customers, contact details, and total order counts generating maximum revenue.

	CustomerID	CompanyName	ContactName	TotalOrdersCount	TotalRevenue
1	QUICK	QUICK-Stop	Horst Kloss	28	117483.39
2	SAVEA	Save-a-lot Markets	Jose Pavarotti	31	115673.39
3	ERNSH	Ernst Handel	Roland Mendel	30	113236.68
4	HUNGO	Hungry Owl All-Night Grocers	Patricia McKenna	19	57317.39
5	RATTC	Rattlesnake Canyon Grocery	Paula Wilson	18	52245.90
6	HANAR	Hanari Carnes	Mario Pontes	14	34101.15
7	FOLKO	Folk och f HB	Maria Larsson	19	32555.55
8	MEREP	Mre Paillard	Jean Fresnere	13	32203.90
9	KOENE	Kniglich Essen	Philip Cramer	14	31745.75
10	QUEEN	Queen Cozinha	Lcia Carvalho	13	30226.10

Question 3: Which categories are most profitable?

- **SQL Techniques:** LEFT JOIN, WHERE, ROUND, ISNULL, UPPER
- **Description:** Analyzed category-wise performance, total orders, units sold, and revenue distribution for active products.

	CategoryID	CategoryName	TotalOrdersCount	TotalUnitsSold	TotalRevenue
1	1	BEVERAGES	316	8407	281744.3500
2	4	DAIRY PRODUCTS	303	9149	251330.5000
3	3	CONFECTIONS	295	7906	177099.1000
4	8	SEAFOOD	291	7681	141623.0900
5	2	CONDIMENTS	185	5000	107893.6000
6	5	GRAINS/CEREALS	154	3865	91394.4000
7	7	PRODUCE	98	2350	78403.0000
8	6	MEAT/POULTRY	69	1658	24633.0000

Question 4: Which regions/countries perform best?

- **SQL Techniques:** Views, Subqueries, GROUP BY, Aggregate Functions
- **Description:** Created dynamic performance views (VW_RegionalPerformance) to track regional sales and order distribution across countries.

	Country	Region	TotalOrders	RegionalUnitsSold	TotalRevenue
3	USA	ID	31	9330	115673.39
4	France	NULL	77	3254	85498.76
5	Brazil	SP	49	4247	60969.30
6	Sweden	NULL	37	2235	59523.70
7	Ireland	Co. C...	19	1684	57317.39
8	Brazil	RJ	34	4247	53999.18
9	USA	NM	18	9330	52245.90
10	UK	NULL	33	2742	40663.71

Question 5: How does performance change over time?

- **SQL Techniques:** CTEs, Window Functions (SUM OVER, LAG), Temporal Functions
- **Description:** Tracked monthly revenue trends, running totals, and month-over-month performance comparisons over time.

	SalesYear	SalesMonth	MonthlyRevenue	RunningTotalRevenue	PreviousMonthRevenue
1	1996	7	30192.10	30192.10	0.00
2	1996	8	26609.40	56801.50	30192.10
3	1996	9	27636.00	84437.50	26609.40
4	1996	10	41203.60	125641.10	27636.00
5	1996	11	49704.00	175345.10	41203.60
6	1996	12	50953.40	226298.50	49704.00
7	1997	1	66692.80	292991.30	50953.40
8	1997	2	41207.20	334198.50	66692.80
9	1997	3	39979.90	374178.40	41207.20
10	1997	4	55699.39	429877.79	39979.90
11	1997	5	56823.70	486701.49	55699.39
12	1997	6	39088.00	525789.49	56823.70
13	1997	7	55464.93	581254.42	39088.00

Question 6: Underperforming Products & Inactive Customers?

- **SQL Techniques:** RIGHT JOIN, Set Operations (EXCEPT), ISNULL
- **Description:** Isolated dead stock/underperforming products and identified inactive customers who have zero recorded orders.

	ProductID	ProductName	TotalUnitsSold	TotalRevenue
3	15	Genen Sho...	122	1813.50
4	67	Laughing L...	184	2562.00
5	74	Longlife Tofu	297	2566.00
6	37	Gravad lax	125	3047.20
7	3	Aniseed Syr...	328	3080.00
8	52	Filo Mix	500	3383.80
9	50	Valkoinen s...	235	3510.00
10	66	Louisiana H...	239	3519.00

	CustomerID	CompanyName	ContactName
1	FISSA	FISSA Fabrica Inter. Salchichas S.A.	Diego Roel
2	PARIS	Paris spcialits	Marie Bertrand
3	Val2	IT	Val2
4	VALON	IT	Valon Hoti

Section 4: Data Warehousing & Analytical Data Modeling

4.1 Analytical Data Model Architecture

To transform raw transactional tables into an enterprise-ready analytical model, a relational data warehouse schema was designed:

- * **Fact Tables:** order_details and All_Orders serve as central Fact tables containing quantifiable metrics (e.g., Sales Amount, Quantity, UnitPrice, and Freight costs).

- * **Dimension Tables:** Contextual entities (customers, products, categories, employees, shippers, territories, region) surround the fact tables to provide filtering and slicing attributes.

- * **Granularity (Grain):** The fact table grain is defined at the individual line-item level per order, enabling detailed micro-level calculations and aggregations.

4.2 Schema Design & Data Warehouse Concepts

- * **Star Schema Implementation:** Applied a classic Star Schema framework featuring single-directional, one-to-many (1:*) relationships extending from Dimension lookup tables to Fact tables. This structure prevents complex circular dependencies and ensures high query performance.

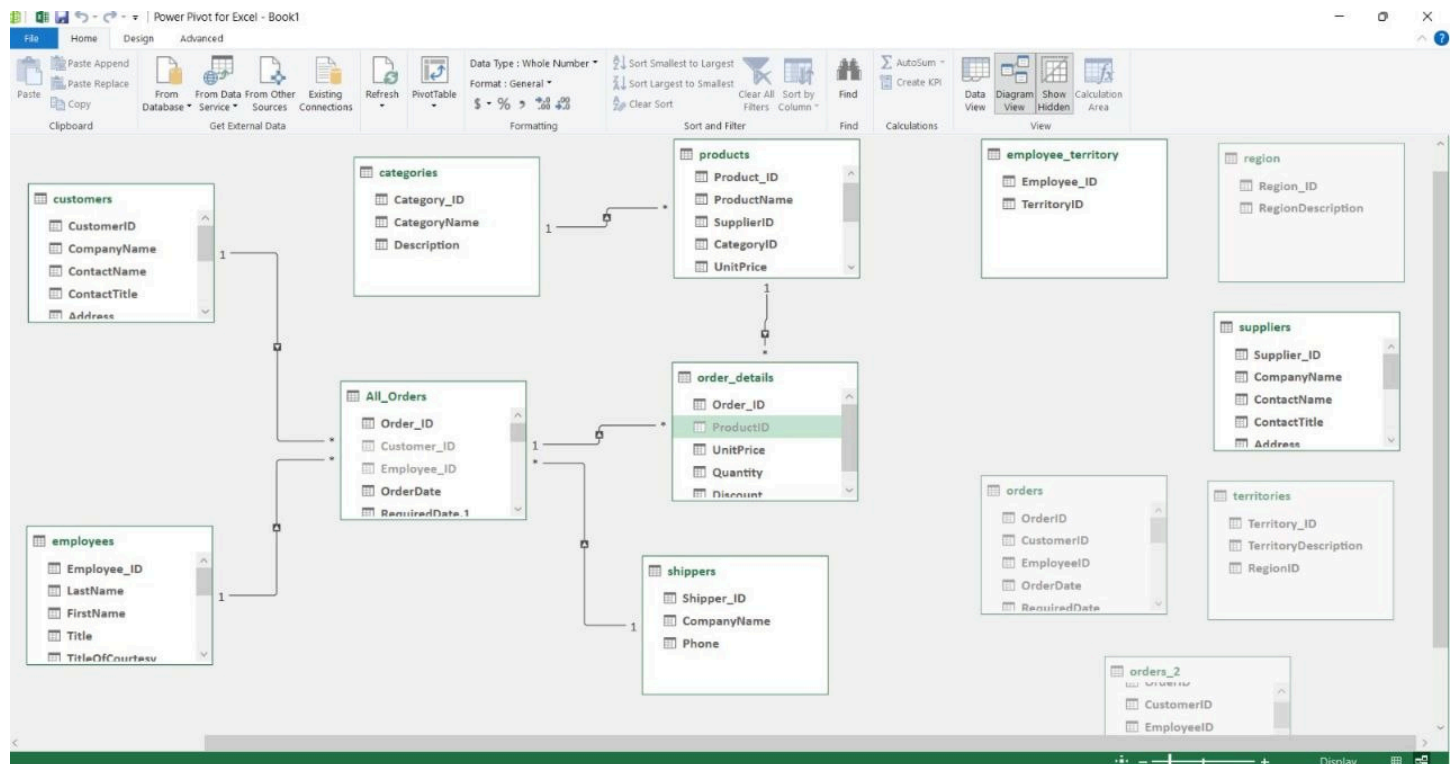
- * **Multi-Source Data Ingestion (ETL):** Integrated multiple source formats (1996–1997 Excel historical records and

1998 CSV files) through Power Query, transforming and unifying them before loading into the analytical engine.

4.3 Data Model Optimization

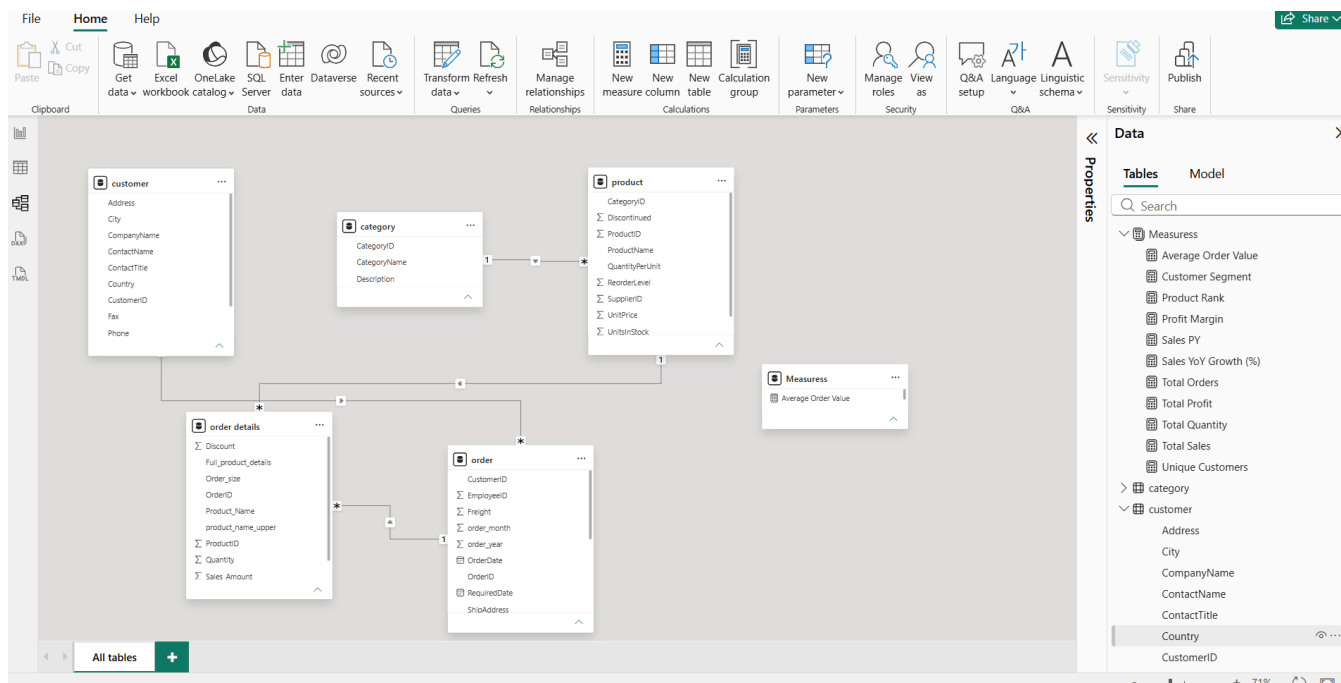
*** Model Cleanup (Hiding Keys):** Primary and Foreign Key columns (such as CustomerID, EmployeeID, and ProductID) were set to "Hide from Client Tools".

*** Business Rationale:** Key columns are required purely for relational joining logic. Hiding them reduces visual clutter, optimizes memory usage, and ensures end-users only interact with descriptive textual attributes (e.g., Customer Name, Product Name).



Section 5: Power BI – Business Intelligence Dashboard

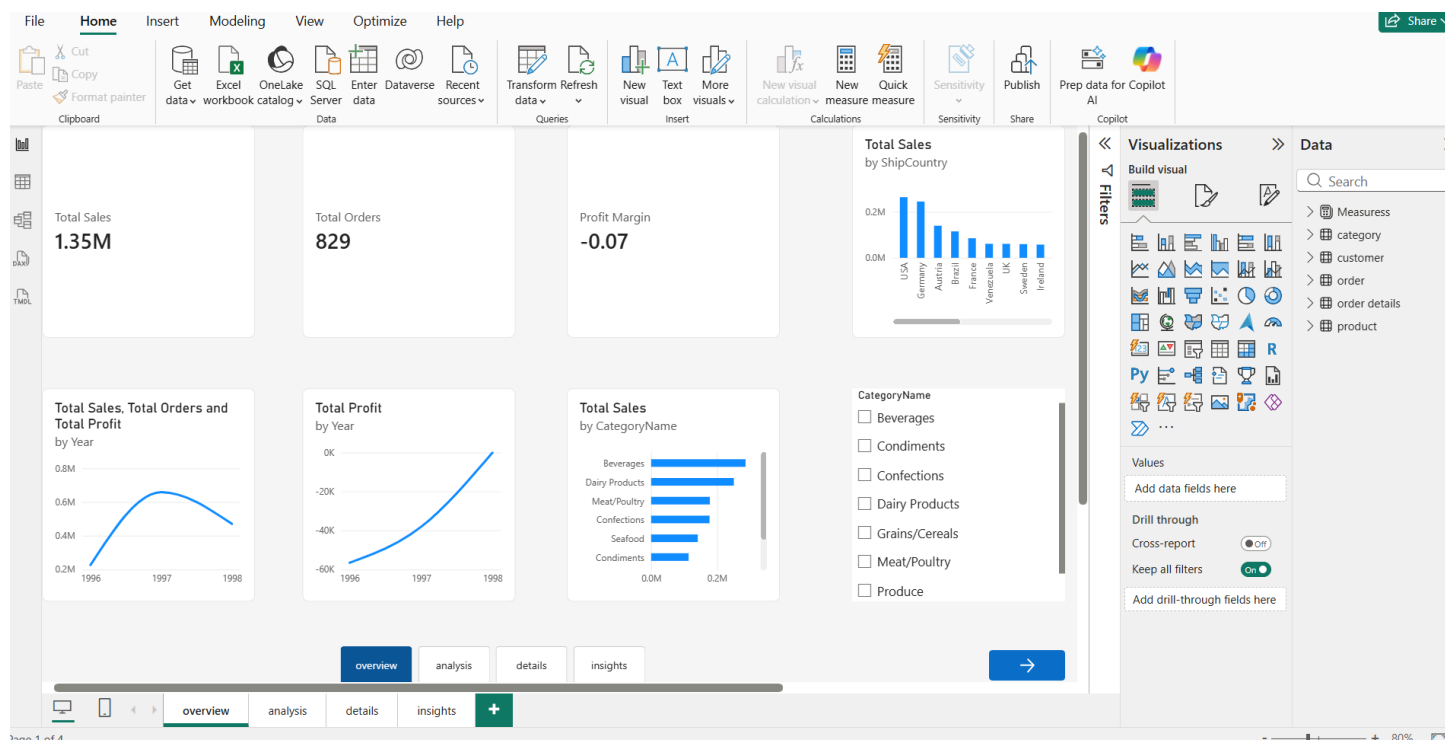
5.1 Data Modeling & DAX Measures Setup



Technical Documentation:

- **Data Model Architecture:** Validated the **Star Schema** relationships (\$1:*\$) connecting the Fact tables (order, order details) with Dimension tables (customer, product, category).
- **DAX Centralization (Measures Table):** Created a dedicated, clean Measure Table named **Measures** to centralize all core DAX formulas:
 - **Core KPIs:** Total Sales, Total Profit, Total Quantity, Total Orders, Unique Customers, Average Order Value, and Profit Margin.
 - **Advanced Analytics:** Configured Customer Segment, Product Rank, Sales PY, and Sales YoY Growth (%).
- **Model Optimization:** Foreign key columns were hidden from client view to maintain an uncluttered workspace.

5.2 Page 1 — Executive Overview

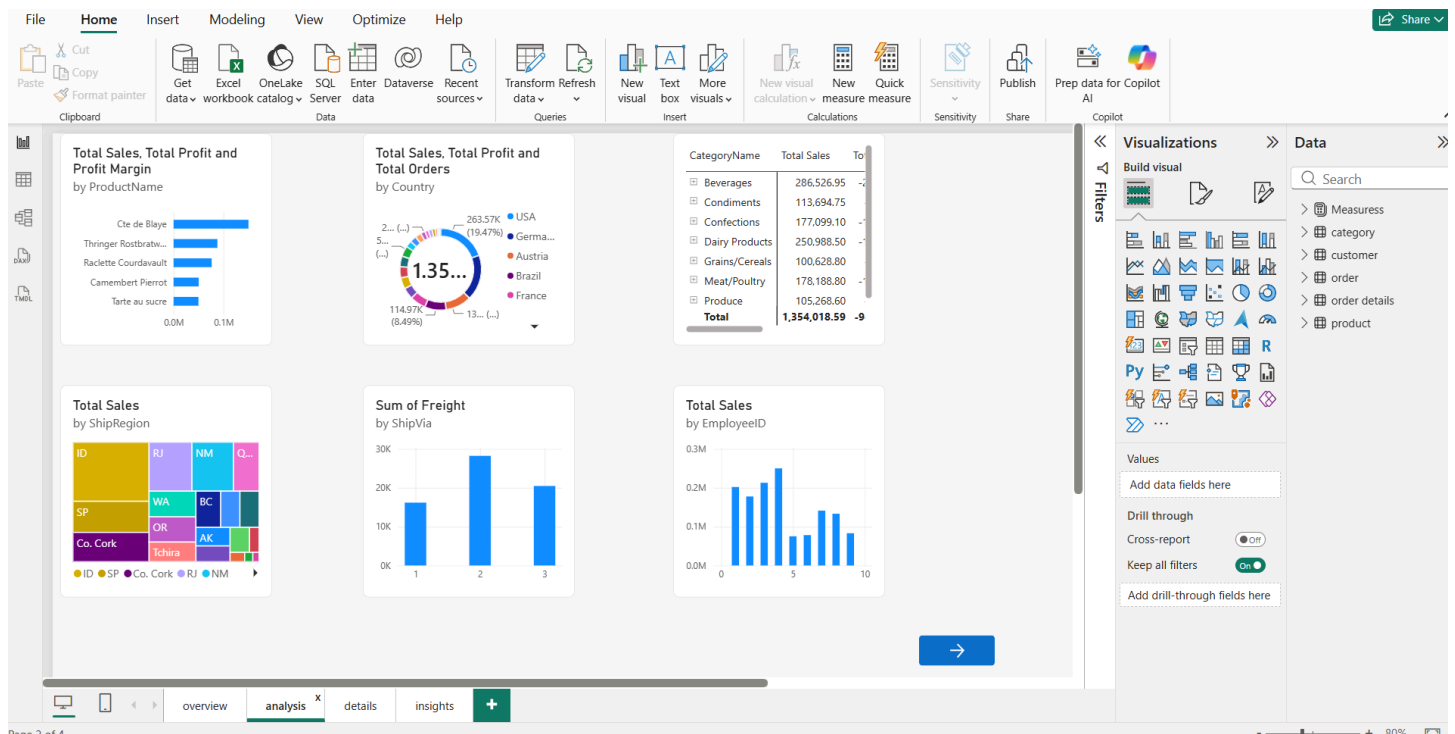


Technical Documentation & Business Insights:

- **Executive KPI Cards:** Displayed top-level business health indicators:
 - **Total Sales:** \$1.35M
 - **Total Orders:** 829
 - **Profit Margin:** -0.07 (Highlights critical overall profitability challenges).
- **Trend & Performance Analysis:**
 - **Sales & Profit Trends:** Combined line and column visuals showing sales and profit trajectories across 1996, 1997, and 1998.
 - **Regional Performance:** Bar chart depicting Total Sales by ShipCountry identifying top geographic markets (led by USA and Germany).

- **Category Breakdown:** Horizontal bar chart displaying sales performance across product categories (led by Beverages and Dairy Products).
- **Interactivity & Date Filtering:** Integrated interactive Slicers (Category Name list) and applied a page-level date filter covering the complete historical range (excluding 4 July 1996). Included interactive page navigation buttons at the bottom.

5.3 Page 2 – Detailed Business Analysis



Technical Documentation & Interactivity:

- **Multi-Dimensional Business Analysis:**
 - **Product & Category Performance:** Horizontal bar chart evaluating sales vs. profit margins by product.
 - **Geographic Distribution:** Donut chart illustrating sales distribution across top countries (USA representing 19.47% / 263.57K).
 - **Regional Heatmap:** Treemap visual displaying sales density across regional markets (ShipRegion).
 - **Logistics Performance:** Column chart analyzing shipping costs (Sum of Freight) across shipping methods (ShipVia).
 - **Sales Representative Coverage:** Column chart tracking sales volume generated by individual employees (EmployeeID).
- **Matrix Visual with Drill-Down:** Implemented a hierarchical Matrix visual (CategoryName $\rightarrow$ ProductName). Users can click the expand (+) icons to drill down from high-level categories into individual product performance.

5.4 Page 3 — Drill-Through Transaction Details

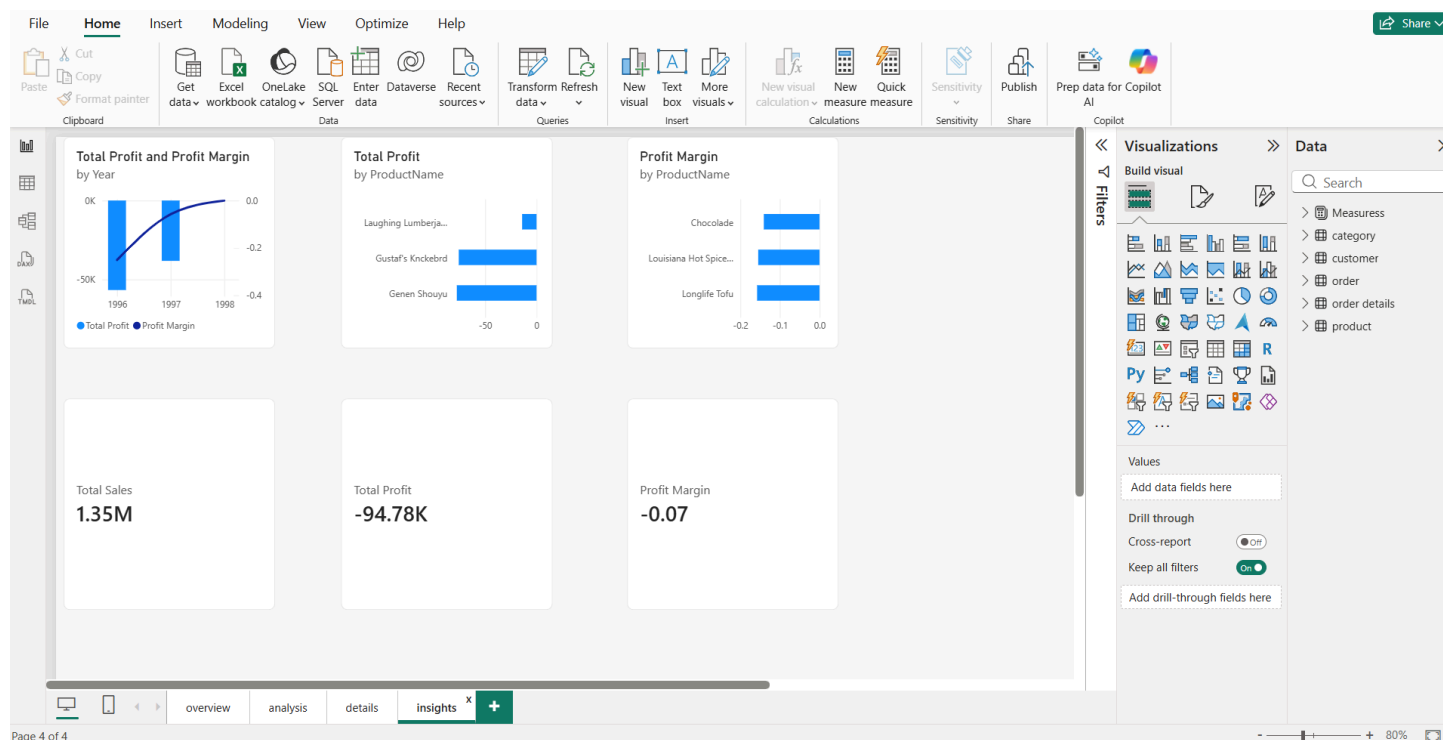
The screenshot displays the Microsoft Power BI Desktop interface. The main canvas shows a table of transaction details. The table has five columns: ProductName, OrderID, Total Sales, Total Profit, and Sum of Freight. The data is filtered by Country = USA. The table is titled 'details' and is the active view in the bottom navigation bar. The right-hand pane shows the 'Visualizations' and 'Data' panes. The 'Visualizations' pane shows a 'Table' visual selected. The 'Data' pane shows a search for 'Country' and the result 'USA'.

ProductName	OrderID	Total Sales	Total Profit	Sum of Freight
Alice Mutton	10294	468.00	-117.00	147.26
Alice Mutton	10338	624.00	-156.00	84.21
Alice Mutton	10346	1,123.20	-280.80	142.08
Alice Mutton	10415	62.40	-15.60	0.20
Alice Mutton	10564	624.00	0.00	13.75
Alice Mutton	10607	3,900.00	0.00	200.24
Alice Mutton	10696	780.00	0.00	102.55
Alice Mutton	10714	1,053.00	0.00	24.49
Alice Mutton	10852	234.00	0.00	174.05
Alice Mutton	10861	1,638.00	0.00	14.93
Alice Mutton	11064	3,003.00	0.00	30.09
Aniseed Syrup	11077	40.00	0.00	8.53
Boston Crab Meat	10482	147.00	-37.00	7.48
Boston Crab Meat	10574	36.80	0.00	37.60
Boston Crab Meat	10607	772.80	0.00	200.24
Boston Crab Meat	10714	1,053.00	0.00	24.49
Total		263,566.98	-17,171.40	13,771.29

Technical Documentation:

- **Target Drill-Through Page:** Configured as a dedicated detail view. When users right-click a specific entity (e.g., country USA) from Page 2 and select **Drill-Through**, it redirects directly to Page 3.
- **Granular Audit Table:** Displays detailed transactional records filtered by the selected context, showing ProductName, OrderID, Total Sales, Total Profit, and Sum of Freight.
- **User Experience:** Includes an automated **Back Button** ($\leftarrow$) in the top-left corner for smooth dashboard navigation back to the primary analysis page.

5.5 Page 4 — Performance & Key Insights



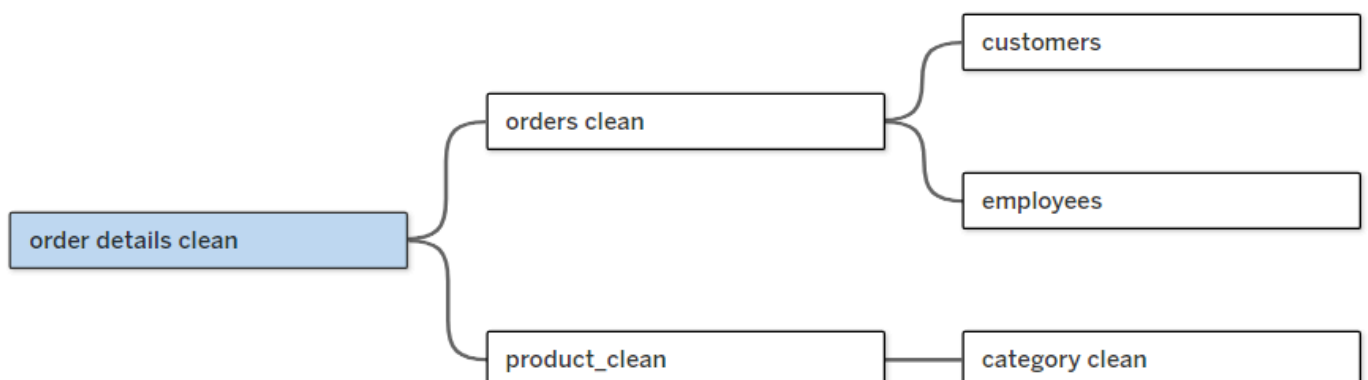
Technical Documentation & Key Findings:

- **Underperforming Area Identification:**
 - Bar charts highlighting severe profit loss areas (Total Profit by ProductName and Profit Margin by ProductName).
 - Flagged key underperforming products generating negative returns (e.g., Laughing Lumberjack, Gustaf's Knäckebröd, Chocolate).
- **Financial Summary Cards:** Summarized total bottom-line financial impact:
 - **Total Sales:** \$1.35M
 - **Net Total Profit:** -\$94.78K (Severe net profit deficit driven by high freight costs and negative-margin items).
 - **Net Profit Margin:** -0.07 (-7%).

Section 6: Tableau – Interactive Data Visualization

6.1 Data Connection & Organization

- **Data Source & Integration:** Connected the consolidated multi-sheet Excel data source directly into Tableau.
- **Data Organization:** Configured multi-table relationships and logical joins across core entities (orders clean, order details clean, customers, employees, product_clean, and category clean).
- **Data Preparation:** Managed multiple measures, dimensions, and data types to ensure clean exploratory visualization.



6.2 Dashboard Design & Visualization Types

Dashboard Layout: Designed a comprehensive, executive-ready interactive dashboard featuring multiple coordinated visual

components:

KPI Cards: High-level metric tracking for Total Sales (\$1,354.46K) and Total Orders (830).

Bar Charts: Category Sales distribution and Top Products ranking based on revenue.

Line Charts: Sales Trend over time, broken down by quarters and years.

Geographic Mapping: Region Map visualizing sales performance and geographic distribution.

Total Sales

\$1,354.46K

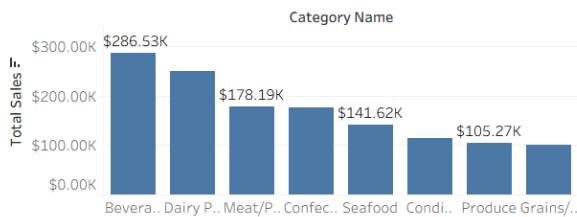
Total Sales

Total Order

830

Total orders

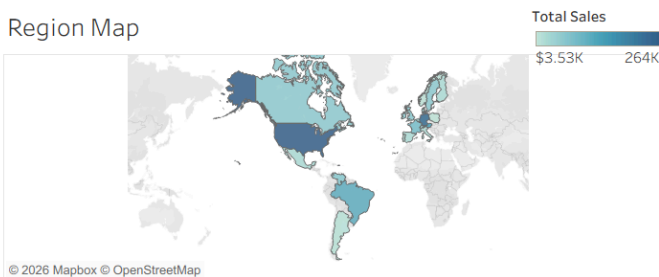
Category Sales



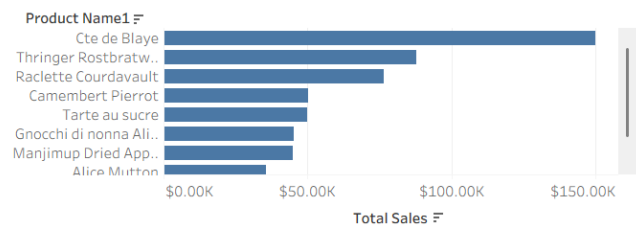
Sales Trend



Region Map



Top Products



6.3 Interactivity & Filtering

- **Interactive Filtering:** Implemented global filters (such as Order Year and interactive dashboard actions) enabling users to dynamically slice, dice, and filter data across all connected views simultaneously.
- **Exploratory Analysis:** Allowed seamless visual exploration of the Northwind dataset, driving clear business insights into product performance, regional trends, and category profitability.

Section 7: Business Insights & Recommendations

6.1 Key Business Findings & Answers to Project Questions

1. High-Performing Revenue Drivers:

Top Product: Cte de Blaye is the primary single revenue generator (\$149,984.20).

Top Category: Beverages is the most commercial category, accounting for \$281,744.35 in revenue across 316 orders.

Top Clients: A concentrated group of key accounts (QUICK-Stop, Save-a-lot Markets, and Ernst Handel) generate over \$346,000 in combined sales, highlighting strong core client retention.

2. Underperforming Areas & Critical Operational Deficits:

Lowest Generating Revenue Items: Products such as Genen Shouyu (\$1,813.50), Laughing Lumberjack (\$2,562.00), and Longlife Tofu (\$2,566.00) demonstrate minimal demand and low revenue conversion.

Profitability & Freight Deficit (Crucial Finding): Despite total sales reaching \$1.35M, net profit yields a severe overall loss (-\$94.78K, Profit Margin: -0.07).

Root Cause: Analysis indicates unoptimized freight and logistics expenses across shipping channels (ShipVia) combined with steep unit discounting, which erodes gross profit margins on high-volume items.

3. Historical Sales Trends & Growth Trajectory:

Sales exhibited steady upward monthly momentum starting from July 1996 (\$30,192.10) through early 1997, peaking in January 1997 at \$66,692.80 before stabilizing around ~\$55K monthly.

6.2 Strategic Recommendations for Management

- **Freight Cost Restructuring:** Re-negotiate carrier contracts (ShipVia) or re-evaluate the shipping fee structure charged to clients. Current high freight overhead directly converts profitable sales into net losses.

- **Product Portfolio Rationalization:** Discontinue or re-price chronic loss-making products (Laughing Lumberjack, Chocolate, Genen Shouyu). Shift marketing focus and inventory expenditure toward high-margin drivers (Beverages & Dairy Products).
- **Key Account Loyalty & Retention:** Create VIP incentives and long-term contracts for top clients (QUICK-Stop, Save-a-lot Markets, Ernst Handel) to secure high-volume predictable cash flow.
- **Dynamic Discounting Guardrails:** Implement strict DAX/system thresholds preventing sales reps from applying discounts that reduce transaction profit margins below zero.